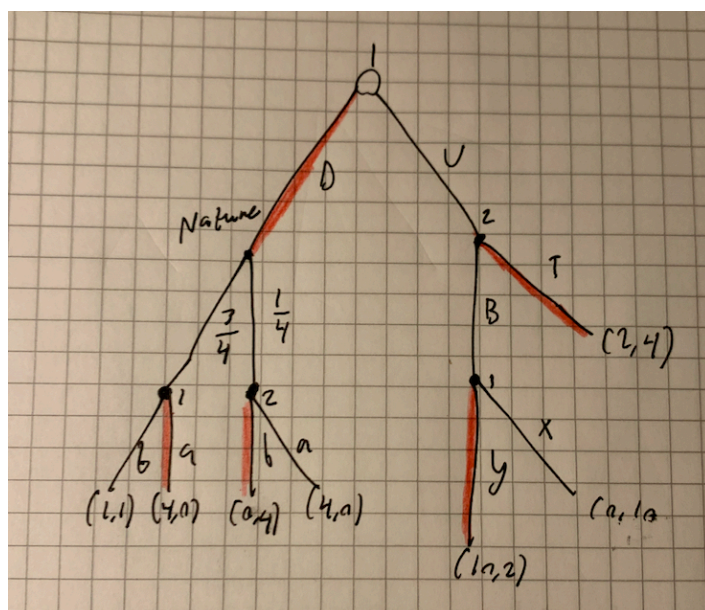
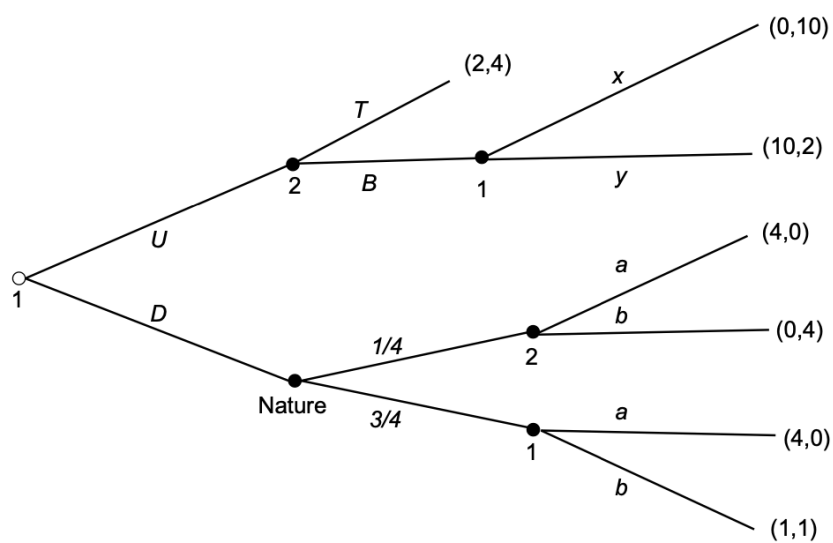


TØ 3 Mikroøkonomi 2 – 2024

Problem 1

Use backwards induction to solve the following game:



Backwards induction works like this:

First, I look at the nodes furthest to the right, i.e., (0, 10) and (10, 2).

Player 1 chooses here. He can either get a utility of 0 in x or 10 in y so of course he chooses y and so on.

Expected nature utility of player 1 will be

$$E[N] = \frac{1}{4} \cdot 0 + \frac{3}{4} \cdot 4 = \frac{12}{4} = 3$$

So, he/she will choose D

Subgame-perfect NE is

$$SPNE = \{(p_1, p_2)\} = \{(D \ y \ a); (T \ b)\}$$

Problem 2

Tom and Jerry are engaged in a test of bravery: They are driving two trucks loaded with dynamite toward each other on a narrow, straight road. Neither is willing to yield.

1

-
- If both veer away, their utility is 0 (resulting in a stalemate).
 - If one veers and the other does not, the driver who swerves faces a utility of -1 (losing the test of bravery), while the other driver's utility is 1 (winning the test of bravery).
 - If neither driver veers, it ends in a collision, resulting in a utility of -10 for both.
- (a) Represent this game in normal form and find all Nash equilibria, i.e., in pure and mixed strategies.
- (b) Now consider previous situation with the twist that Tom's steering wheel can be detached. Tom thus has the choice of throwing his steering wheel out of the window before he and Jerry start driving towards each other and decide (simultaneously) whether to veer or not. Assume that if Tom throws his steering wheel out, this will be observed by Jerry.
- b1) Model this game as a sequential game, i.e., provide the game tree.
 - b2) List Tom's and Jerry's pure strategies.
 - b3) Find all pure-strategy subgame perfect Nash equilibria.
 - b4) Write down the normal form of the game and describe one Nash equilibrium which is not subgame perfect.
 - b5) What should Tom do?

a.

First of all, Tom is player 1 and Jerry 2 since later in the problem, Tom has the opportunity to throw his wheel, making him the first player to move

v is veer and s is stay

By BR, the PSNE are $PSNE = \{(s, v), (v, s)\}$

AD	v	s
-v	0, 0	-1, 1
s	1, -1	-10, -10

And mixed

$$0q - 1(1 - q) = 1q - 10(1 - q) \Leftrightarrow q^* = \frac{9}{10}$$

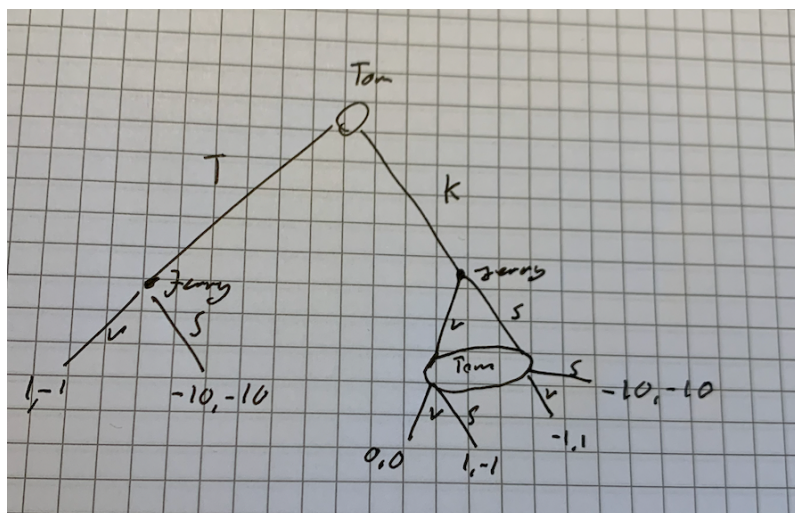
$$0p - 1(1 - p) = 1p - 10(1 - p) \Leftrightarrow p^* = \frac{9}{10}$$

Meaning that

$$MSNE = \left\{ \left(\frac{9}{10}, \frac{1}{10} \right), \left(\frac{9}{10}, \frac{1}{10} \right) \right\}$$

		q	1-q
p	AD	v	s
1-p	-v	0, 0	-1, 1
	s	1, -1	-10, -10

b.1



Like before, T is throw, K is keep, v is veer and s is stay

b.2

Tom's pure strategies:

$$\{Tv, Ts, Kv, Ks\}$$

Jerry's

$$\{vv, vs, sv, ss\}$$

b.3 and b.4

Probably easier to first write up the normal form, find the NE and then the SPNE

$\begin{array}{c} 2 \\ 1 \end{array}$	vv	vs	sv	ss
Tv	1, -1	1, -1	-10, -10	-10, -10
Ts	1, -1	1, -1	-10, -10	-10, -10
Kv	0, 0	-1, 1	0, 0	-1, 1
Ks	1, -1	-10, -10	1, -1	-10, -10

By BR, there are clearly seven NE. Note that (Kv, vs) is not a NE, I made a mistake when underlining

$\begin{array}{c} 2 \\ \swarrow \\ 1 \end{array}$	vv	vs	sv	ss
Tv	<u>1, -1</u>	<u>1, -1</u>	-10, -10	-10, -10
Ts	<u>1, -1</u>	<u>1, -1</u>	-10, -10	-10, -10
Kv	0, 0	<u>-1, 1</u>	0, 0	<u>-1, 1</u>
Ks	<u>1, -1</u>	-10, -10	<u>1, -1</u>	-10, -10

The NE are

$$NE = \{(Tv, vv), (Tv, vs), (Ts, vv), (Ts, vs), (Kv, ss), (Ks, vv), (Ks, sv)\}$$

The SPNE are

$$SPNE = \{(Tv, vs), (Ts, vv), (Ks, vv)\}$$

Why are these SPNE?

Look at the subgame after player 1 has chosen K

Two equilibria here: player 2 choosing v and player 1 choosing s since player 2 will not deviate: $-1 > -10$. The other is player 2 choosing s and player 1 choosing v. Player 2 will not deviate since: $1 > 0$

So, every SPNE must have s on second place of player 1's strategy and v on second place of player 2's strategy OR v on player 1's second strategy and s on player 2's second strategy:

$$(xs, vv) \text{ or } (xv, vs), x \in \{T, K\}$$

b.5

Throwing his steering wheel will leave him with a payoff of 1 since Jerry will swerve

Keeping it, we will have the game from part a where he will get

$$0 \cdot pq + (-1) \cdot p \cdot (1 - q) + 1 \cdot (1 - p) \cdot q + (-10) \cdot (1 - p) \cdot (1 - q) = -0.1$$

Since $p = q = 0.9$

		q		1-q
		V		S
P	AD	-V	0, 0	-1, 1
1-P	S	1, -1	-10, -10	

Since Tom's payoff is greater from throwing his steering wheel away, he should do that

Problem 3

Line-item veto. In American politics the President can veto a budget passed by Congress. Ronald Reagan once argued for a “line-item veto” that would allow him to eliminate parts of a budget bill presented. His argument was that Congress had a tendency to put in so-called pork-barrel expenditures, spending that pleases the constituency of particular members of the Congress but is not in the national interest. With the line-item veto, he argued, he could strike such items out that would never survive if presented individually rather than as part of a budget with other items.

- (a) Do you agree with the proposition that the freedom to veto parts of a bill can only increase the president’s power and never yield any worse outcomes from his perspective? Write down your intuition.
- (b) Suppose there are two items of expenditure under consideration: urban renewal (U) and an antiballistic missile system (M). Congress likes U, the President likes M, but both prefer a package of U and M over the status quo. Congress moves first to propose a bill and chooses between $\{U + M, U, M, \text{status quo}\}$. The president moves next, if a bill is submitted. The following table gives the payoffs of possible scenarios to both players:

Outcome	Congress	President
Both U and M	3	3
U only	4	1
M only	1	4
Neither (status quo)	2	2

- b1) Draw a game tree for the situation where the president has no line-item veto. That is, he can only decide to sign the budget bill or veto it and get the status quo. What is the SPNE outcome?
- b2) Draw a game tree for the situation where the president has a line-item veto. That is, he can decide to sign the budget bill, strike out a single item, or veto the entire bill. What is the SPNE outcome?
- b3) Is the president (Congress) better or worse off with the line-item veto power than without it?

a.

It could yield a worse outcome for the president and the congress if the congress and the president have very different views.

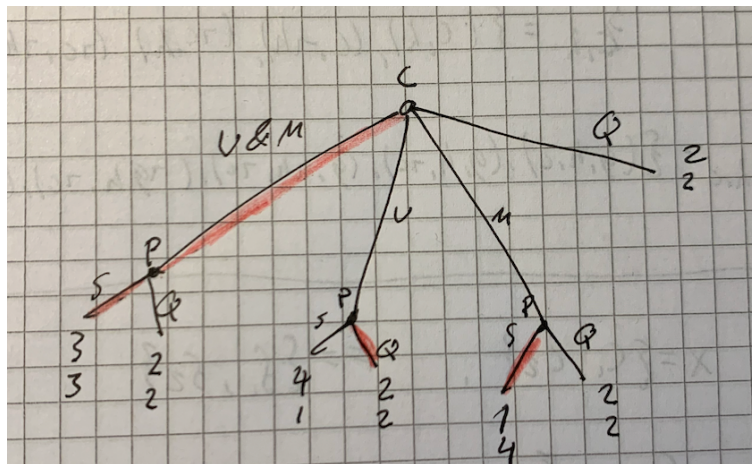
b.1

C: Congress

P: President

S: Sign

Q: Status quo



Congress will never choose status quo nor M. Therefore, I only focus on U&M and U. The president will sign if the Congress choose U&M and veto it if Congress chooses U only. This means that the outcome will be U&M.

$$SPNE = \{(U\&M, SQS)\}$$

b.2

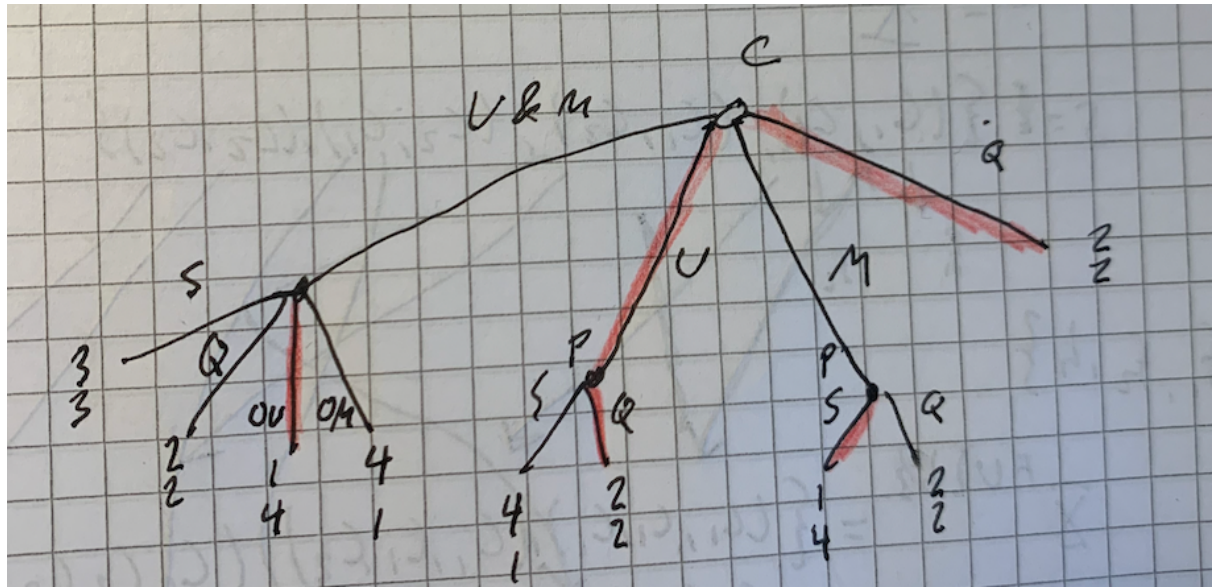
OU: Out U

OM: Out M

Now, the president can rule out one thing

This shows that congress will in this situation choose status quo or propose U and get it rejected. No matter what, it will lead to the status quo outcome:

$$SPNE = \{(Q, OU \text{ } Q \text{ } S), (U, OU \text{ } Q \text{ } S)\}$$

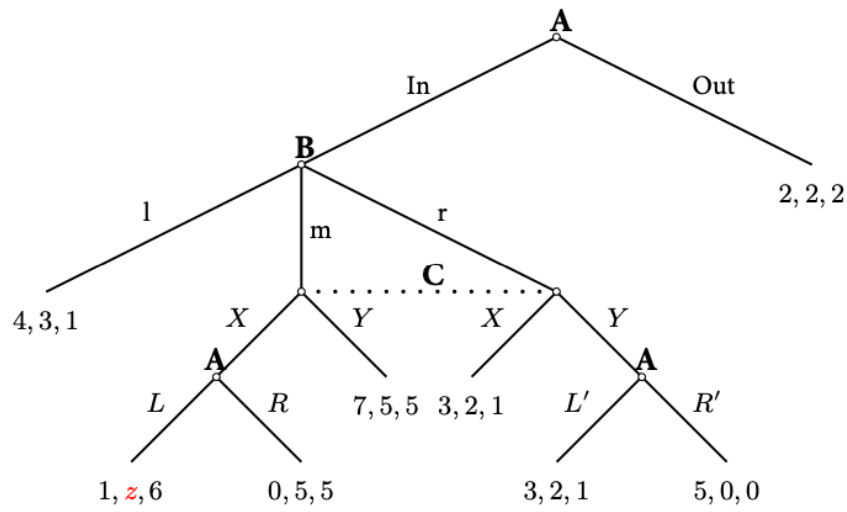


b.3

Both parties are worse off since they now have a payoff of 2 instead of 3

Problem 4

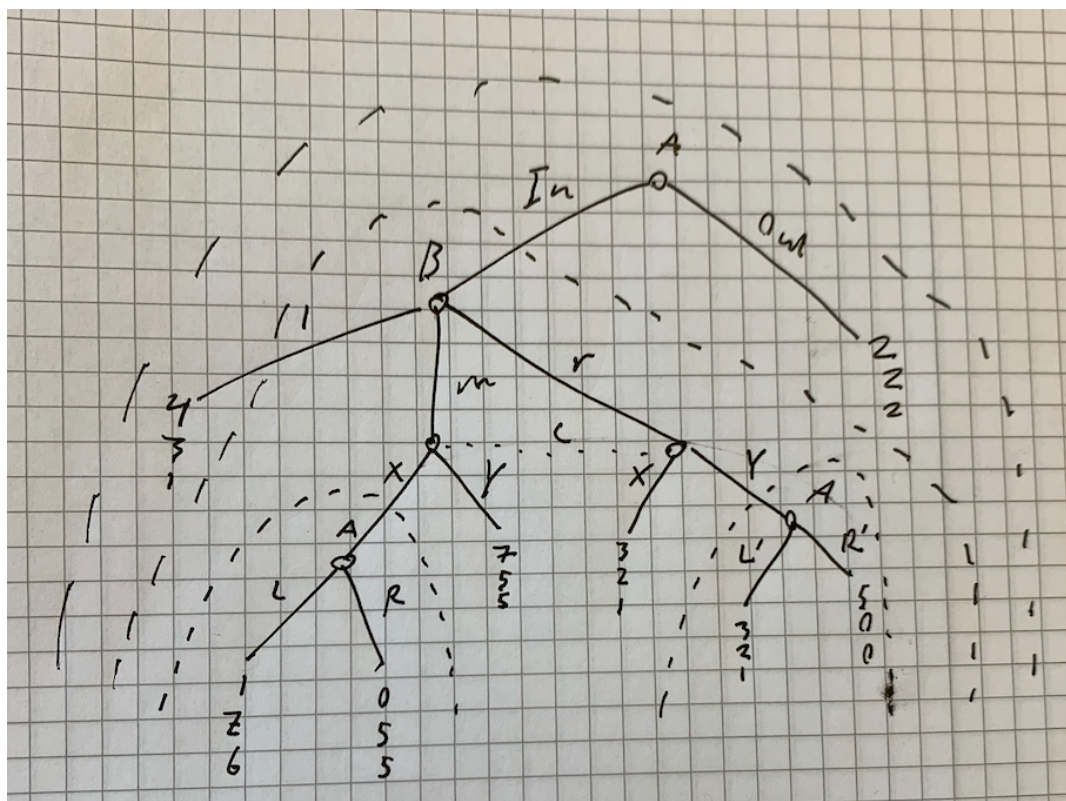
Consider the following extensive form game between players **A**, **B** and **C**:



- How many subgames does this game have (also counting the entire game as a subgame)?
- Write down all pure strategies of players **A**, **B** and **C**.
- Find all pure strategy subgame perfect Nash equilibria.

a.

There are four subgames. Games with imperfect information are not considered as subgames:



b.

Player A has eight pure strategies:

S_A

$= \{(In, L, L'); (In, L, R'); (In, R, L'); (In, R, R'); (Out, L, L'); (Out, L, R'); (Out, R, L'); (Out, R, R')\}$

Player B has three pure strategies:

$$S_B = \{(l), (m), (r)\}$$

Player C has two pure strategies because of uncertainty

$$S_C = \{(X); (Y)\}$$

c.

Start at the end nodes, i.e., Player A's choices:

- In the node to the left, he can choose a utility of either 1 or 0, meaning that he chooses L
- In the node to the right, he can choose a utility of 3 or 5, implying that he chooses R'

For player 3, Y is strictly dominated by X:

in the scenario to the left, choosing X will leave player 3 with a payoff of 6. Choosing Y will give her 5

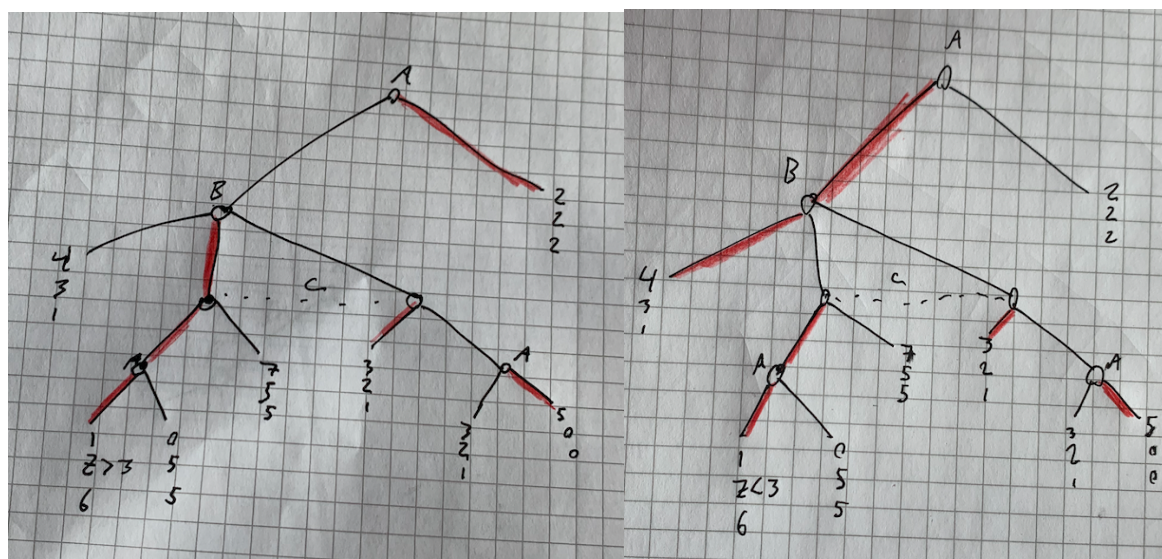
To the right, choosing X will give her 1 and choosing Y will give her 0 since player A chooses R'

Player B can get a payoff of 3 by choosing l . If he instead chooses m , he will get z payoff.

Notice that choosing r will never happen since here, he will get a payoff of 0

So, player B will choose m if $z > 3$. If $z = 3$, he is indifferent

If player B chooses m , player A will choose Out because $2 > 1$ and if player B chooses l , player A will choose In because $4 > 2$



In symbols:

$$SPNE = \{(Out, L, R'); (m); (X)\}, \quad z > 3$$

$$SPNE = \{(In, L, R'), (l); (X)\}, \quad z < 3$$

And when $z = 3$, B is indifferent between strategy l and m , so we have two SPNE

$$SPNE = \{(Out, L, R'); (m); (X)\} \cup \{(In, L, R'), (l); (X)\},$$